Teaching notes and solutions

> **Teacher note · Teacher notes.** Begin with the three expressions of the dot product. Orthogonal lines are not necessarily intersecting: "perpendicular" requires an intersection. To prove that a line is orthogonal to a plane, choose two non-collinear directions. The general point-to-plane distance formula is not required by heart: favour the orthogonal projection. The Cartesian equations used in the S54 progression can be introduced through the vector condition, then formalised in S67.

Solutions to the activity

1. A point H of d is written $H = A + t\vec{u}$, so $H(1+t, t, 0)$ for $t \in \mathbb{R}$.
2. $\overrightarrow{MH} = (1+t, t-2, -3)$. The dot product with $\vec{u} = (1,1,0)$ is $1+t+t-2 = 2t-1$. Setting it to zero gives $t = \frac{1}{2}$ and $H(\frac{3}{2}, \frac{1}{2}, 0)$.
3. The frame is orthonormal: $MH^2 = (\frac{3}{2})^2 + (-\frac{3}{2})^2 + (-3)^2 = \frac{27}{2}$. So $MH = \frac{3\sqrt{6}}{2}$.
4. For $P \in d$, $\overrightarrow{HP}$ is collinear with $\vec{u}$ and hence orthogonal to $\overrightarrow{MH}$. Pythagoras' theorem gives $MP^2 = MH^2 + HP^2 \geq MH^2$, with equality only if $P = H$. The point found is indeed the unique point of the line closest to M.

E1. In the orthonormal frame, the dot product is $2 \times 1 + (-1) \times 4 + 3 \times 0 = -2$.

E2. The dot product is $3a + 2(-1) + 1 \times 4 = 3a + 2$. Orthogonality is equivalent to its vanishing: $3a + 2 = 0$, so $a = -\frac{2}{3}$.

E3. The two direction vectors are non-collinear and both dot products are zero. The midpoint is $I(1,1,1)$. The required locus is the plane through I with normal vector $\overrightarrow{AB} = (2,2,2)$, hence with normal vector $(1,1,1)$.

E4. Both vectors are non-zero, with norms $\sqrt{2}$, and their dot product is 1. So their angle $\theta \in [0, \pi]$ satisfies $\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$, that is $\theta = \frac{\pi}{3}$.

E5. The area of the right-angled base is $\frac{3\times 4}{2} = 6\,\text{cm}^2$. The height perpendicular to this base is 5 cm, so $V = \frac{6\times 5}{3} = 10\,\text{cm}^3$.

Solutions to HW11

1. $\overrightarrow{AB} = (-2,2,0)$, $\overrightarrow{AC} = (-2,0,2)$ and $\overrightarrow{BC} = (0,-2,2)$; the three norms are $2\sqrt{2}$. The triangle is equilateral. Its height is $\sqrt{(2\sqrt{2})^2 - (\sqrt{2})^2} = \sqrt{6}$; its area is therefore $\frac{(2\sqrt{2})(\sqrt{6})}{2} = 2\sqrt{3}$.
2. The vectors $\overrightarrow{AB}, \overrightarrow{AC}$ are non-collinear. Their dot products with $(1,1,1)$ are respectively $-2+2 = 0$ and $-2+2 = 0$. The non-zero vector $\vec{n}$ is therefore normal to (ABC).
3. The projection of O lies on the line through O with direction vector $\vec{n}$, so $H = (t,t,t)$. Its condition for belonging to the plane is $(t-2)+t+t = 0$, that is $t = \frac{2}{3}$. We obtain $H(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$, which satisfies both conditions for an orthogonal projection.

4. $OH = \sqrt{3(\frac{2}{3})^2} = \frac{2}{\sqrt{3}}$. Taking ABC as the base, the volume is $V = \frac{(2\sqrt{3})(\frac{2}{\sqrt{3}})}{3} = \frac{4}{3}$ volume units. Other base: the triangle OAB is right-angled at O, with area $\frac{2\times 2}{2} = 2$; the line (OC) is perpendicular to the plane (OAB), and $OC = 2$. Thus $V = \frac{2\times 2}{3} = \frac{4}{3}$.

Solutions to the textbook or past-paper work

Exercise 23. In the given orthonormal frame, the dot product is the sum of the products of the corresponding coordinates:

$$(-3.5)(-4) + 2 \times 5.5 + (-4)(-3) = 14 + 11 + 12 = 37,$$

$$\frac{2}{3}\left(-\frac{3}{4}\right) + \frac{15}{7}\frac{14}{3} + \left(-\frac{12}{11}\right)\left(-\frac{22}{4}\right) = -\frac{1}{2} + 10 + 6 = \frac{31}{2}.$$

Exercise 27. In each case, the two given vectors $\vec{u}$ and $\vec{v}$ span the plane and are not collinear. A non-zero vector $\vec{w}$ is normal if, and only if, it is orthogonal to each of them.

1. $\vec{w} \cdot \vec{u} = -132 + 312 - 180 = 0$ and $\vec{w} \cdot \vec{v} = -66 + 546 - 480 = 0$. The vector $\vec{w}$ is normal to the plane.
2. $\vec{w} \cdot \vec{u} = 20 + 6 - 26 = 0$, but $\vec{w} \cdot \vec{v} = 18 - 6 - 10 = 2 \neq 0$. It is not normal to the plane.
3. $\vec{w} \cdot \vec{u} = \frac{1}{2} + 6 - 7 = -\frac{1}{2} \neq 0$. This single non-zero product is enough to rule out normality.